

Lecture Notes: Regularization Part 3: Elastic Net Regression

I. The Problem: How to Choose?

This lecture assumes you are familiar with **Ridge (L2)** and **Lasso (L1)** regression.

- **Lasso (L1) Regression** works best when your model contains **a lot of useless variables**. It performs feature selection by shrinking these parameters to **zero** [00:43].
- **Ridge (L2) Regression** works best when **most of the variables are useful**. It shrinks all parameters *asymptotically* toward zero but never makes them exactly zero [01:12].

The Problem: What if you have a model with millions of parameters (common in deep learning)? [01:48] You can't possibly know in advance which variables are useful and which are useless.

How do you choose between Lasso and Ridge?

II. The Solution: Elastic Net Regression

The good news is **you don't have to choose** [02:13]. You can use **Elastic Net Regression**.

Elastic Net sounds fancy, but it is simply a **hybrid** that combines both the Lasso and Ridge penalties [02:29].

- **Least Squares** minimizes:

$$SS_{Residuals}$$

- **Elastic Net** minimizes:

$$SS_{Residuals} + \lambda_1 \sum |\beta_i| + \lambda_2 \sum \beta_i^2$$

It is simply the sum of the **Lasso (L1) penalty** [02:38] and the **Ridge (L2) penalty** [02:42].

III. How Elastic Net Works

By combining both penalties, Elastic Net gets the **best of both worlds** [04:41].

1. Two Tuning Parameters (λ)

The model now has *two* λ values to tune, one for each penalty:

- λ_1 controls the strength of the Lasso (L1) penalty.
- λ_2 controls the strength of the Ridge (L2) penalty.

We use **cross-validation** to find the best *combination* of λ_1 and λ_2 [03:07].

2. A Flexible Hybrid

By adjusting the two λ values, Elastic Net can morph into the other models:

- If $\lambda_1 > 0$ and $\lambda_2 = 0$:
The Ridge penalty disappears, and the model is **pure Lasso Regression** [03:24].
- If $\lambda_1 = 0$ and $\lambda_2 > 0$:
The Lasso penalty disappears, and the model is **pure Ridge Regression** [03:35].
- If $\lambda_1 = 0$ and $\lambda_2 = 0$:
Both penalties disappear, and the model is **standard Least Squares** [03:14].
- If $\lambda_1 > 0$ and $\lambda_2 > 0$:
The model is a **true hybrid** that balances the strengths of both Lasso and Ridge [03:44].

IV. The "Superpower" of Elastic Net: Correlated Variables

Elastic Net has a specific, powerful advantage that makes it better than Lasso or Ridge on their own: **it handles correlated parameters (variables) very well** [03:53].

- **The Problem:** Imagine you have two variables that are highly correlated (e.g., **Weight** and **BMI**).
- **Lasso's Behavior:** When faced with a group of correlated variables, Lasso tends to **arbitrarily pick just one** of them and set the others to zero [04:03]. This isn't ideal, as it loses information.
- **Ridge's Behavior:** Ridge tends to **shrink all the correlated variables together**, keeping all of them in the model [04:11].
- **Elastic Net's Behavior:** By combining the two, Elastic Net gets the best of both. It **groups the correlated variables** (like Ridge) and then either **keeps all of them** in the model or **removes all of them at once** (like Lasso) [04:20].

This "grouping" behavior is much more stable and provides better, more reliable models when dealing with highly correlated features.

V. Summary

1. **Elastic Net** combines the **Lasso (L1) penalty** and the **Ridge (L2) penalty**.
2. It uses two tuning parameters, λ_1 and λ_2 , which are found using cross-validation.
3. It gets the "best of both worlds":
 - It can **perform feature selection** (like Lasso).
 - It can **handle useful parameters** (like Ridge).
4. Its biggest advantage is **handling groups of correlated variables** more effectively than either Lasso or Ridge alone.

http://googleusercontent.com/youtube_content/10